Kestrel Ridge Technologies Ltd. - Material Properties (FICTIONAL)

1. Structural and shaft steels

Grade	Yield strength (MPa)	Tensile strength (MPa)	Elongation (%)	Density (kg/m³)
S275	275	430	23	7850
S355	355	470	22	7850
42CrMo4	650	900	12	7850
AISI 316	205	515	40	8000

2. Allowable stresses for coupling design

Component	Material	Allowable stress (MPa)	Safety factor
Hub	42CrMo4	430	1.5
Shaft	S355	236	1.5
Key	C45	180	2.0